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The reaction between some dyes and synthetic hydroxyapatite
1. *The uptake of dyes in relation to their structures*

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Synopsis. The uptake of dyes from dilute solutions by synthetic hydroxyapatite and other sparingly soluble calcium compounds has been determined. About 30 dyes, mostly azo-, dis-azo and anthraquinonoid types were used in 95% ethanol or 0.1 M tris buffer. Many had closely related configurations. Chemical groupings possibly responsible for the adsorption of particular dyes by hydroxyapatite have been deduced from an analysis of the results. The uptake of most dyes from alcoholic solutions was linearly related to the surface area of hydroxyapatite. Calcium carbonate and secondary calcium phosphate took up less stain than hydroxyapatite of similar surface area. With the simpler anthraquinonoid dyes, the uptake was higher from aqueous than alcoholic solutions, but specificity for hydroxyapatite was much less. The increased uptake of dye by powdered bone or dentine when rendered anorganic was proportional to the increased surface area. It was found that several dyes in common use as stains for bone and calcified tissue were only poorly adsorbed by synthetic hydroxyapatite under the particular conditions of these experiments.

The experimental data presented could be used as a basis for the development of histochemical reactions for calcified tissue or inclusions. By suitable choice of dyes, solvent and rinsing solution it ought to be possible to differentiate various forms of calcified material.

Introduction

In a study of the maturation of hypomineralized areas of rat enamel, stains were used to follow the progressive increase in the degree of mineralization (Speirs, 1967). While searching for suitable dyes, it was found that some, though quite closely related chemically, varied considerably in their ability to stain immature enamel. This was attributed to the inability of certain stains to penetrate the enamel or to their failure to be adsorbed on to the hydroxyapatite crystals or possibly on to the protein matrix. It was primarily with a view to settling this issue that some preliminary investigations were planned.

Although many staining methods are described in histochemical texts for the localization of mineralized tissues and calcified deposits within soft tissues, relatively little is known about the reactions involved (Ercoli & Lewis, 1943; Conn, 1961; Baker, 1958; McGee Russell, 1958; Kiel & Heertjes, 1963; and Myers, 1964). The vast literature on dyeing

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chemistry gives little help with this question. This dearth of data seemed to justify an extension of the preliminary studies to include a more detailed investigation of both the relationship between the chemical structure of dyes and their ability to stain synthetic hydroxyapatite and also the nature of the mechanism by which the stains bind to mineralized tissue. These two studies will be presented separately.

It was anticipated that an analysis or comparison of the behaviour of many different dyes towards hydroxyapatite and other calcium salts might reveal some patterns which would enable relationships to be deduced between the structure of a dye and its staining properties.

It was considered that this information might be of use in the design of histochemical reactions in studies in which the chemical nature of calcified material was of interest. It was hoped that the decision to use a particular dye for the localization and possible identification of the calcified tissue could be supported by experimental data concerning the specificity of the dye and the nature of the staining reaction. The limitations of applying the results obtained in the present study to the situation in histological reactions are, however, substantial.

Throughout this report, the meaning of 'adsorption' may cause confusion. No precise definition is intended; the word is used loosely and no distinction is being made between chemical combination or 'ionic adsorption' and non-ionic, physical adsorption.

Several factors determined the selection of dyes for study. Stains mentioned in the literature as suitable for treating calcified inclusions and for revealing bone structure were listed (Conn, 1961; Gurr, 1960). Where possible, dyes with closely related chemical structures were also obtained so that the effects of relatively minor rearrangements or substitutions within the molecules on staining properties could be examined. Other dyes were included whose formulae suggested the possibility of chelate-ring formation with calcium in hydroxyapatite. Again dyes showing small variations of the parent formulae were also used. Apart from a few miscellaneous stains, most could be grouped into azo-, dis-azo and anthraquinonoid types (Table 1).

Table 1. List of dyes used, classification, molecular weights, formulae and reasons for inclusion

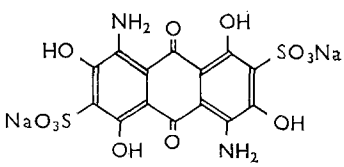
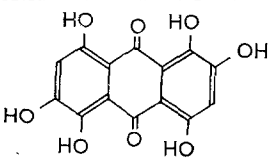
Dye	*Colour index No.*[1]	*Molecular weight*	*Feature of interest*
Anthracene derivatives – anthraquinonoids			
Acid Alizarin Blue GR 	63015	506	degree of substitution and position of groups
Acid Alizarin Blue SWR 	58605	304	for comparison with Alizarin Bordeaux

Table 1 (*contd*)

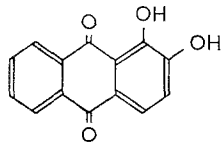
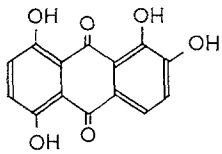
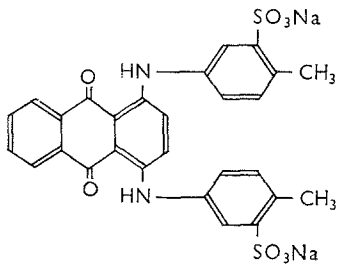
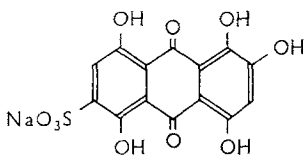
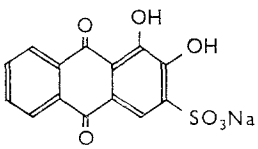
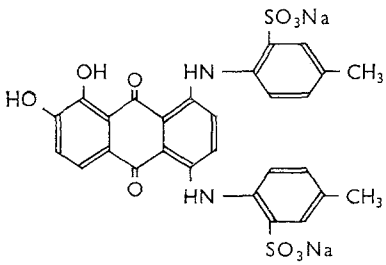
Dye	*Colour index No.*[1]	*Molecular weight*	*Feature of interest*
Alizarin 	58000	240	parent molecule; possible chelate ring formation with calcium (Myers, 1964)
Alizarin Bordeaux 	58500	272	for comparison with Alizarin; used as histochemical reagent for magnesium
Alizarin Cyanin Green 	61575	622	used for staining bone (Gurr, 1960)
Alizarin Cyanin 2R 	?58520	390	for comparison with Alizarin Bordeaux
Alizarin Red S 	58005	342	used extensively for staining bone (Gurr, 1960; Myers, 1964)
Alizarin Viridin 	62555	654	for comparison with Alizarin Cyanin Green and Alizarin

Table 1 (*contd*)

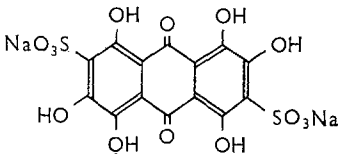
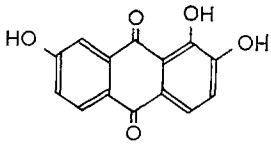
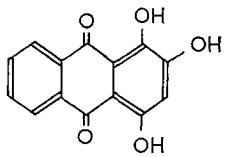
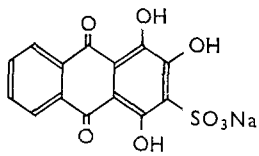
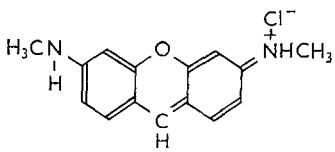
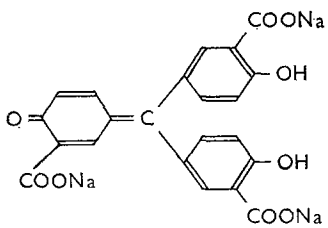
Dye	*Colour index No.*[1]	*Molecular weight*	*Feature of Interest*
Anthracene Blue SWX 	58610	508	for comparison with Alizarin Cyanin 2R
Anthrapurpurin 	58255	256	feebly acid dye used to demonstrate calcium deposits (Gurr, 1960)
Purpurin 	58205	256	used to demonstrate insoluble calcium salts in cells (Gurr, 1960)
Purpurin Sulphonate 	58210	253	for comparison with Purpurin and Alizarin Red S
Miscellaneous dyes			
Acridine Red 	45000	274	used for staining deposits of calcium salts
Chrome Violet G 	43810	488	formula suggests possible chelate formation

Table 1 (*contd*)

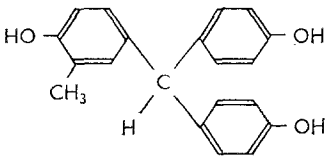
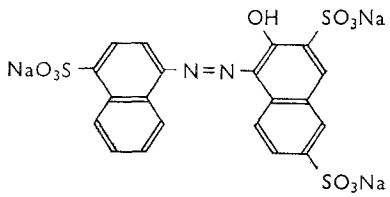
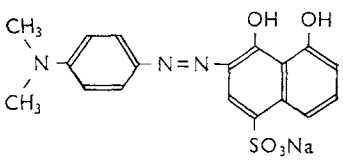
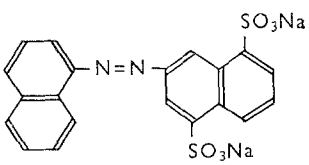
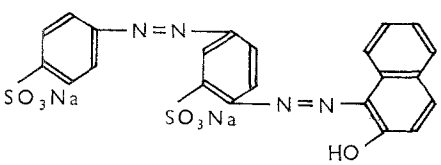
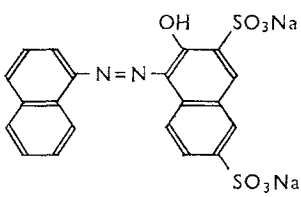
Dye	*Colour index No.*[1]	*Molecular weight*	*Feature of interest*
Rosolic Acid 	43800	295	for comparison with Chrome Violet G
Azo and dis-azo dyes			
Amaranth 	16185	604	for comparison with Bordeaux Red and Ponceau 2R
Azo Acid Blue B 	16545	409	for comparison with Orseillin BB, Chromotrope and others
Azo Bordeaux 	15020	502	for comparison with Bordeaux Red; Conn (1961) gives this name as synonym of Bordeaux Red
Biebrich Scarlet 	26905	556	in conjunction with Methylene Blue used for differential staining of bone and cartilage; for comparison with Orange II
Bordeaux Red 	16180	502	

Table 1 (contd)

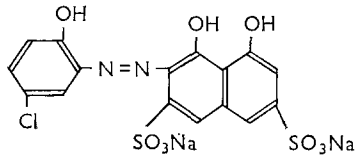
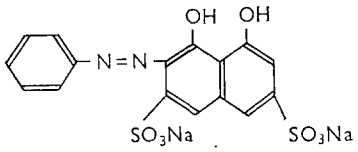
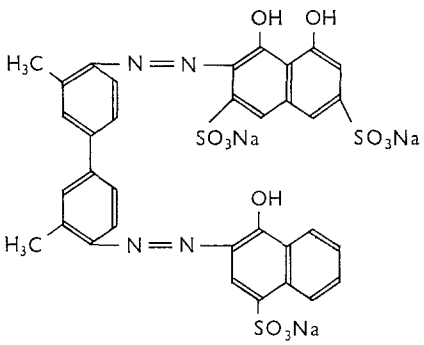
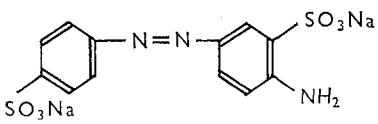
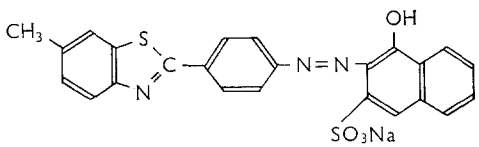
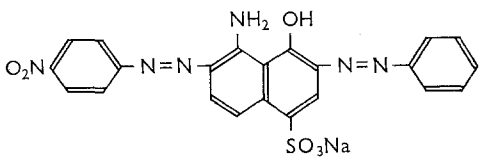
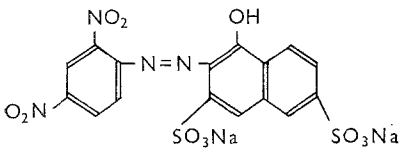
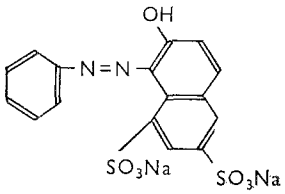
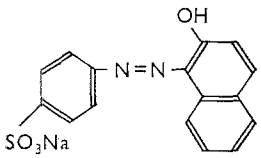
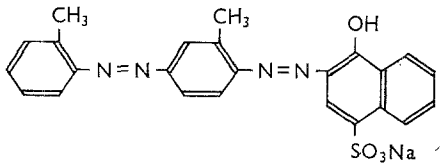
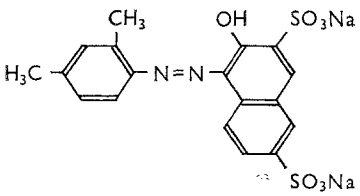
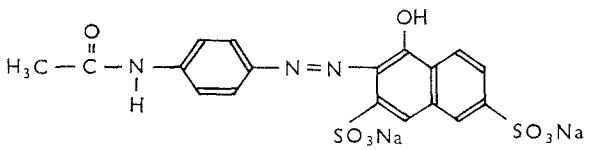
Dye	*Colour index No.*[1]	*Molecular weight*	*Feature of interest*
Calcium Fast Purple 	16680	519	forms lakes with inorganic calcium salts
Chromotrope 2R 	16570	468	for comparison with Calcium Fast Purple
Dianil Blue 	23690	855	for influence of two terminal groups seen in Orseillin BB and Calcium fast purple; largest mol. wt
Fast Yellow 	13015	401	used for staining sections of bone (Conn, 1961)
Geranine G 	14930	497	for comparison with similar end grouping in Orseillin BB and Dianil Blue
Naphalene Black 10B 		516	for comparison with group in Orseillin BB; capable of staining immature enamel (Speirs, 1967)

Table 1 (*contd*)

Dye	*Colour index No.*[1]	*Molecular weight*	*Feature of interest*
Nitrazine Yellow 	14890	542	for comparison with Sorbine Red
Orange G 	16230	452	used as stain for embryonic bone (Gurr, 1960)
Orange II 	15510	350	simple structure in which to study azo group–hydroxyl relationship
Orseillin BB 	26670	584	
Ponceau 2R 	16150	480	for comparison with Amaranth and other closely related dyes
Sorbine Red 	14895	509	for comparison with Geranine G

[1] Colour index nos. are those quoted in Conn (1961) and in the Colour Index Series published in 1956 by the British Society of Dyers and Colourists and the American Association of Textile Chemists and Colourists.

Materials and methods

Four samples of hydroxyapatite were used. They were manufactured as 'calcium phosphate, dentifrice grade' (Albright & Wilson Ltd, England, and Monsanto Chemical Co., St. Louis, Missouri). They have been well characterized as to their composition, limits of contaminants, surface area and diffraction patterns. There is now evidence to show that these hydroxyapatite samples contain quantities of β tricalcium phosphate and that they have Ca : P ratios slightly less than theoretical. Secondary calcium phosphate ($CaHPO_4.2H_2O$) and calcium carbonate were obtained commercially (British Drug Houses Ltd, Poole, England) and fluorapatite was supplied in two geological forms by Griffin & George Ltd, England, and R. D. Parkinson, Shepton Mallet, England. The determinations of surface area of the hydroxyapatite samples were kindly carried out by Unilever Ltd, while those of the other substances used in this work were undertaken by the Paint Research Station, Teddington, Middlesex, and by Mr J. D. W. Hudson of Imperial College of Science and Technology, London. In all these determinations, the B.E.T. nitrogen gas adsorption method was used. The hydroxyapatite samples had surface areas of 68, 36, 25 and 8 m^2/g; secondary calcium phosphate 1.9; calcium carbonate 0.6; and fluorapatite 1.79. All the stains were supplied by G. Gurr Ltd and E. Gurr Ltd, London. A complete list of these is given in Table 1 with their formulae, molecular weights, colour index numbers and reasons for inclusion. No attempt was made to improve the degree of purity of any dye. It was considered appropriate to use the dyes as they would be used for histochemical and routine histological work. However, with some of the dyes the influence of contaminating salts on the uptake of the dyes by hydroxyapatite was determined. The method used was to dialyze dye solutions in tris buffer against several changes of tris buffer through 20/32 Visking cellophane dialysis tubing. The sodium and potassium concentrations of the dialysate and original dye solution were determined using an EEL flame photometer. The adsorption of the dye by hydroxyapatite after

Table 2. Contamination of dyes by sodium and potassium

Dye	*Observed total %K*	*Observed total % Na*	*Calculated % Na as contaminant (total observed −theoretical)*
Alizarin Bordeaux	<1.0	0	0
[1]Alizarin Cyanin 2R	0	12.5	6.6
[1]Alizarin Red S	0	20.2	13.5
[1]Anthracene Blue SWX	0	15.8	6.7
Calcium Fast Purple	—	17.2	8.2
Dianil Blue	0	26.5	18.4
[1]Geranine G	0	30.7	26.1
Orseillin BB	<1.0	16.3	12.4
Ponceau 2R	<1.0	16.8	7.2
[1]Purpurin Sulphonate	0	1.6[2]	−5.1

[1] The affinity of these dyes for hydroxyapatite was studied before and after dialysis.

[2] This is less than the formula demands, and would suggest that there might be an admixture of Purpurin, or that the sulphonate is not present as the sodium salt.

the salt contaminants had been eliminated in this way was compared with that of the original dye solution. With the five dyes studied (indicated in Table 2), there was no consistent or significant difference in the percentage uptake of dye between the two forms of the dye solutions. The sodium and potassium contaminations of ten dyes are shown in Table 2. It is of interest to note that King & Pruden (1968) have recently reported large quantities of sodium and potassium as chlorides and sulphates in some samples of Alizarin Red S, but found only negligible amounts of organic impurities. They estimated the free salt content of their four samples to be from 23 to 52%. Assuming that all the sodium in our sample of Alizarin Red S is present either as chloride or sulphate this would be equivalent to a salt content of about 34 or 42%, respectively.

Absorption spectrograms were prepared for each dye solution before and after exposure of the solution to hydroxyapatite. In this way the uptake of dye was determined at the wavelength characteristic of the stain and any heterogeneity in the dye solution was noted. In those dyes which gave absorption peaks with subsidiary shoulders, *e.g.* Alizarin Viridin, there was no evidence of preferential uptake of any minor component of the solution by hydroxyapatite. After exposure to hydroxyapatite the pattern of the spectrogram was normally identical in shape with that of the original solution, but in some instances the peak was displaced by 10–20 nm after exposure to hydroxyapatite. Such a finding was observed with some dyes in aqueous solution, but rarely with those in alcoholic solutions. It was attributed to the formation of a coloured soluble complex of calcium or phosphate with the dye.

It is known that many dyes will form molecular aggregates or micelles in solution, particularly at high concentrations. This tendency is reduced in the presence of lower alcohols (Kipling, 1965) and in solutions of low ionic strength. The non-linear relationship between optical density and solute concentration (*e.g.* with Ponceau 2R) was attributed to this effect. These and other considerations led to the decision to prepare all the dyes at concentrations of less than 0.6 mM in either 95 : 5 ethanol : water, or in 0.1 M tris-chloride buffer, pH 7.4–7.5. Variations in pH or ionic strength have not been studied. All solutions were filtered before use through Whatman No. 42 paper in a millipore filter funnel. Even at these low concentrations difficulty was experienced in dissolving some of the anthraquinonoid dyes in tris buffer. Calibration curves were plotted for all dye solutions.

In preliminary studies three experimental variables were examined. When variations in the duration of the exposure of hydroxyapatite to dye solutions were investigated, it was found that near equilibrium had been established in 15 min of vigorous shaking. The uptake of dyes continued at a very slow rate for several hours. In these experiments, 50 mg hydroxyapatite samples were used in 25 ml 0.4 mM dye solution, the surface area of the hydroxyapatite in these and in routine studies being 36 m^2/g.

The second variable was the weight of hydroxyapatite. It was necessary to use samples of such weight that in the presence of a constant volume and constant initial concentration of dye in solution there would be the maximum variation between different dyes in their uptake. There should also be sufficient difference between the colour intensity of initial and final solutions as to be readily determined by the spectrophotometer. Graphs were drawn in which the uptake of different dyes from 0.2 mM solutions in alcohol was plotted against the weight of the hydroxyapatite samples. One such graph is illustrated in Fig. 1. It was decided that 50 mg samples were suitable. It will be obvious that any ranking of stains according to their uptake by hydroxyapatite is somewhat arbitrary and might be altered by the particular solid : solution ratio being used.

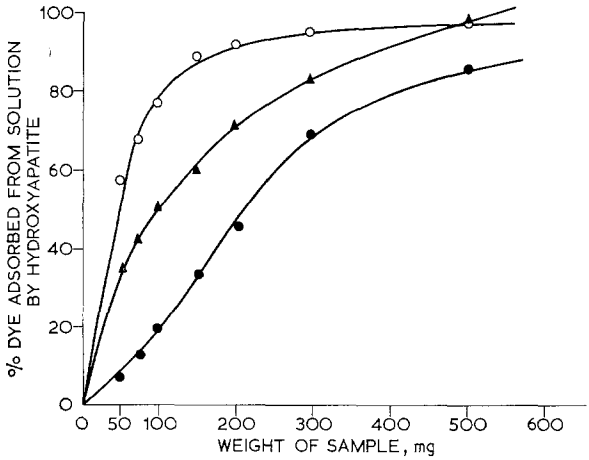


Figure 1. Relationship between the percentage of dye removed from dye solutions and the weight of hydroxyapatite sample exposed to 25 ml dye solution. ○ Orseillin BB; ▲ Chrome Violet; ● Naphthalene Black. Dyes prepared at concentrations of 0.2 mM in 95% ethanol.

The decision to use stains at concentrations of 0.2 mM was also rather empirical. The considerations regarding weight applied equally to the question of dye concentration. A range of initial concentrations between 0.025 and 0.6 mM was studied and the uptake of dye by hydroxyapatite determined. Some typical graphs are shown in Fig. 1 of paper 2 of this series where their significance is discussed more fully.

The uptake of dye was expressed either as μM dye removed from a particular volume of the original solution per gram hydroxyapatite or alternatively as original minus final concentration of dye, as a percentage of the original. Determinations of the colour intensity of the dye solutions were made in a Unicam SP 500 spectrophotometer or an EEL colorimeter. The results shown in the following tables and graphs have not been corrected to take account of the contamination of some of the dyes with free salts.

The uptake of any dye by hydroxyapatite as determined in these experiments gave little information concerning the stability of the binding between stain molecules and solid. Some measure of the degree of retention of the stain by hydroxyapatite was required. The stained hydroxyapatite was therefore exposed to successive 'washings' of aqueous or alcoholic solvents and the intensity of colour imparted to these solutions was determined in the spectrophotometer and compared with standard concentrations of the dye. With some stained hydroxyapatite samples it was possible to remove 50% of the dye during the course of several extractions, while others were much more resistant. The degree of retention of the dyes can be expressed in several ways. In Table 3b the number of washings necessary to extract 50% and 75% of the dye was determined (where possible) and from this was calculated the amount of dye released per 'average' washing expressed as a percentage of the dye taken up initially. However, probably the most meaningful figure is the absolute amount of dye still retained by hydroxyapatite after one 'average' extraction, expressed as micromoles (μM) per gram hydroxyapatite. This composite value provides the best indication of actual uptake and degree of retention.

Results

In general, stain taken up by hydroxyapatite from aqueous dye solutions was well retained during aqueous washing procedures (Table 3a). It was seldom possible to elute 50% of the adsorbed dye. With some notable exceptions, alcoholic washings rapidly removed considerable quantities of the stains previously adsorbed from alcoholic dye solutions (Table 3b). Some dyes are included in only one of these two tables because their uptake by hydroxyapatite from one solvent was negligible.

Several features in these tables deserve comment. It has not been found possible to correlate the differential effects of the two solvents on either uptake or elution of dyes with the differential solubilities of the dyes in water and alcohol quoted by Gurr (1960). The relatively

Table 3a. Uptake and retention of dyes in aqueous solution[1]

Dye	*Mean loss of stain per 'washing' as % of stain taken up initially*	*Mean quantity of dye retained after one extraction μ-moles/g hydroxyapatite*
Orseillin BB	4.5	70.5
Purpurin	6.5	68.5
Alizarin Bordeaux	4.3	66.0
Purpurin Sulphonate	4.5	65.0
Geranine G	4.3	62.0
Alizarin Cyanin 2R	2.5	53.0
Dianil Blue	1.0	50.0
Alizarin	3.0	45.5
Acid Alizarin Blue SWR	13.0	40.0
Anthracene Blue SWX	5.0	40.0
Alizarin Red S	8.0	27.5
Anthrapurpurin	12.0	21.6
Chrome Violet G	0	21.0
Azo Acid Blue	15.0	13.6
Alizarin Viridin	—	12.2[2]
Naphthalene Black	—	10.5[2]
Acid Alizarin Blue GR Alizarin Cyanin Green Azo Bordeaux Calcium Fast Purple Chromotrope Bordeaux Red	—	6.0–10.0[2]
Fast Yellow Sorbine Red Orange II	—	0.5–2.0[2]

[1] These results were obtained with dyes in 0.1 M tris buffer, pH 7.45. The concentration of the dye in solution was 0.2 mM. 50 mg hydroxyapatite were shaken in 25 ml of this solution for 15 min. Complete uptake and retention of dye would correspond to values of 100 μ-moles per gram hydroxyapatite. Each 'washing' consisted of exposure of stained solid (approximately 50 mg) to 5 ml tris buffer, for 15 min.

[2] Calculated before any 'washing'.

Table 3b. Uptake and retention of dyes in alcoholic solution[1]

Dye	*No. of exposures required to desorb stain* 50% *removal*	75% *removal*	*Mean loss of stain per 'washing', as % of stain taken up initially*	*Mean quantity of dye retained after one extraction μ-moles/g hydroxyapatite*
Geranine G	4.6	7.4	10.4	54.0
Orseillin BB	5.0	9.0	8.6	52.7
Bordeaux Red	15.0	—	3.3	51.7
Dianil Blue	—	—	0.4	43.0
Ponceau 2R	5.3	9.4	8.5	35.0
Amaranth	10.3	—	4.8	33.4
Acid Alizarin Blue GR	2.3	3.6	21.2	31.2
Anthracene Blue SWX	1.7	4.0	22.0	29.6
Chrome Violet	—	—	6.0	21.4
Alizarin Viridin	3.6	12.0	8.0	21.2
Naphthalene Black 10B	2.3	3.4	22.0	17.1
Alizarin Cyanin 2R	—	—	6.3	14.9
Alizarin Bordeaux	—	—	7.5	14.8
Orange G	2.6	5.0	16.5	13.7
Alizarin Cyanin Green	2.3	3.6	21.2	13.0
Acid Alizarin Blue SWR	—	—	20.0	12.8
Calcium Fast Purple	1.8	3.4	24.0	10.1
Fast Yellow	1.2	2.6	33.0	9.4
Alizarin Red S	2.7	5.0	16.2	8.4
Azo Bordeaux	1.0	—	42.0	8.0
Chromotrope	0.8	2.0	44.5	6.9
Azo Acid Blue	—	1.4	53.5	5.8

[1] These results were obtained with dyes in 95% ethanol. The concentration of the dye in solution was 0.2 mM. In 25 ml of this solution was shaken 50 mg hydroxyapatite for 15 min. Complete uptake and retention of dye would correspond to values of 100 μ-moles per gram hydroxyapatite. Each 'washing' consisted of exposure of stained solid (approximately 50 mg) to 5 ml ethanol for 15 min.

simple anthraquinonoid dyes, *e.g.* Purpurin and Alizarin, with few substituted groups are adsorbed to a high degree by hydroxyapatite from tris buffer, but not from ethanol. Their solubility is higher in alcohol. The retention of many of these dyes was very great when washed with tris buffer. Purpurin and Alizarin Bordeaux have similar structures (Table 1) and show almost identical adsorption values in tris medium (Table 3a). Further substitution in Alizarin Bordeaux by two hydroxyl groups gives Acid Alizarin Blue SWR and by a sulphonate group and a hydroxyl group gives Alizarin Cyanin 2R. Further substitution gives Anthracene Blue. All these show slightly less uptake than the simpler compounds. The ratings of Alizarin Bordeaux, Purpurin, Purpurin Sulphonate and the dis-azo dye, Orseillin BB may be artificially high and may not in fact indicate the real adsorption of these dyes by hydroxyapatite. With each of these particular dyes, in aqueous solution, two coloured products appeared to form with the hydroxyapatite powder. One of these, considerably darker than the other, was considered to be due to the formation of a colloidal lake with free calcium in solution. These lakes probably became more particulate on standing and sedimented along with the hydroxyapatite during centrifuging. Small variations in time

between the end of the exposure period and centrifuging might account for variations in the apparent uptake of dye seen on different occasions. It was found that the variation in the results between replicate experiments with any one of these dyes was greater than with other dyes. It is worth noting at this stage that the formation of these additional coloured products was also seen when calcium salts other than hydroxyapatite were exposed to these dyes (Table 4).

Of the dis-azo and azo dyes in tris buffer, the greatest uptake by hydroxyapatite was shown by Orseillin BB and Geranine G, with Dianil Blue next. It is seen (Table 1) that identical structural units exist in the dis-azo dyes, Orseillin BB and Dianil Blue, with a very similar unit in the azo dye, Geranine G. At one end of the large Dianil Blue molecule, joined by an azo-linkage, is a naphthalene nucleus with two substituted hydroxyl and two sulphonate groups. This particular unit is also found as the principal part of Chromotrope and Calcium Fast Purple. From the low uptake of these two dyes it seems unlikely that this part of the molecule is playing a major role in the adsorption of Dianil Blue. It is apparent from the dyes listed in Table 1 and the results for their adsorption (Table 3) that the majority of the azo dyes show considerably less affinity for hydroxyapatite in tris buffer than in alcohol.

Chrome Violet G, with a solubility in water ten times that in alcohol, was remarkable in its complete stability when adsorbed to hydroxyapatite and washed with tris solvent. The uptake of this dye was the same in tris and ethanol (Table 3b). There are, however, few other points of similarity between the behaviour of dyes in the two solvents.

In the anthraquinonoid series the simpler members like Alizarin and Purpurin exhibit no measurable uptake by hydroxyapatite in alcohol. Alizarin Bordeaux and Alizarin Cyanin 2R are the least substituted molecules to have an uptake and stability of significance. In fact, the retention factor of these dyes was the highest of all in the anthraquinonoid series. When the structure of Purpurin Sulphonate, which is not adsorbed from alcoholic solution, is compared with that of Alizarin Cyanin 2R it can be appreciated that the degree of substitution is critical. Further substitution produced acid Alizarin Blue GR and Anthracene Blue SWX. These two dyes are more soluble in alcohol than the others and show greater adsorption but lower retention values. It is of interest that in alcohol the amino groups in Acid Alizarin Blue did not exert the inhibitory effect on adsorption seen in tris buffer. It is these groups which distinguish this dye from Anthracene Blue. No attempt was made at this stage to remove the adsorbed dye with a solvent other than that used in the initial treatment of the hydroxyapatite.

With the azo and dis-azo dyes it is again seen that the same three members, Geranine G, Orseillin BB and Dianil Blue, although differing in structure, size and molecular weight, but having one unit in common, are ranked closely in order of uptake by hydroxyapatite. As in the aqueous solvent, Dianil Blue is adsorbed less than might be anticipated from the possession of two functional units, one at each end of the molecule. However, one of these units, present in Chromotrope 2R and Calcium Fast Purple, is again associated with a low order of uptake. These two dyes, which have very similar structures, are adsorbed to an almost identical degree, 12.5 and 13.3 μM/g respectively. The apparent difference seen between them originates from the poorer retention of Chromotrope. The three closely related dyes, Bordeaux Red, Ponceau 2R and Amaranth, are adsorbed to a similar degree. Merely the introduction of a sulphonate group into the naphthalene molecule in Bordeaux Red presumably accounts for the reduced uptake of Amaranth, though the mechanism involved is obscure.

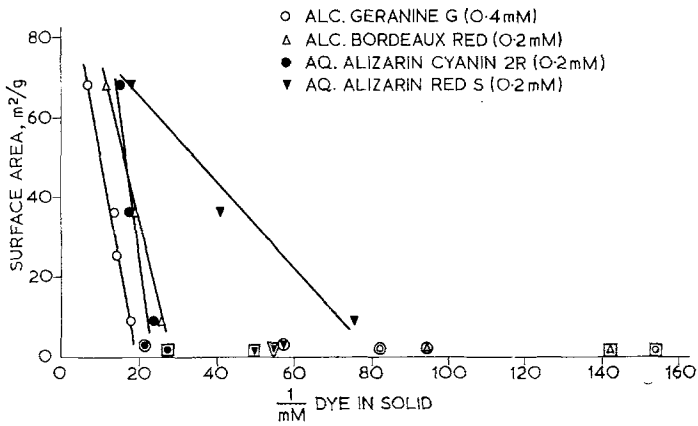


Figure 2. The uptake of dye solutions by hydroxyapatite, secondary calcium phosphate, calcium carbonate and fluorapatite in relation to the surface areas of these solids.
○ 0.4 mM alc. Geranine G; △ 0.2 mM alc. Bordeaux Red; ● 0.2 mM Alizarin Cyanin 2R in tris; ▼ 0.2 mM Alizarin Red S in tris. Large square surround – calcium carbonate; circle surround – calcium phosphate; triangle surround – fluorapatite.

Table 4a. Uptake of dyes (μ-moles) by hydroxyapatite, calcium phosphate, calcium carbonate and fluorapatite, in relation to the surface areas of these compounds

Dye	*Hydroxyapatite*			*$CaHPO_4$*	*$CaCO_3$*	*Fluorapatite*
	68	*36*	*8*	*1.9*	*0.6*	*1.79 m^2/g*
Aqueous						
Acid Alizarin Blue GR	—	9.5	—	8.0	8.0	—
Acid Alizarin Blue SWR	—	40.0	—	12.0	14.0	19.0
Alizarin	53.0	47.0	—	45.0	36.0	—
Alizarin Bordeaux	—	66.0	—	53.5	48.7	74.0
Alizarin Cyanin Green	—	8.8	—	7.2	7.2	—
Alizarin Cyanin 2R	74.0	56.5	39.0	49.0	35.0	—
Alizarin Red S	54.0	25.0	13.3	17.8	17.6	20.0
Alizarin Viridin	—	15.0	—	8.0	10.0	—
Anthracene Blue SWX[1]	56.0	42.0	26.0	22.5	17.0	17.0
Azo Acid Blue	—	16.0	—	15.0	15.0	—
Calcium Fast Purple	—	8.6	—	9.0	14.0	—
Chrome Violet G	—	21.0	—	6.0	1.0	—
Dianil Blue	—	78.0	—	54.0	54.0	—
Geranine G	80.0	53.5	30.5	18.0	21.0	—
Naphthalene Black 10B	—	10.5	—	13.5	10.5	—
Orseillin BB	93.5	92.0	90.0	97.5	97.0	90.0
Ponceau 2R	—	14.0	—	14.0	14.0	—
Purpurin	85.0	73.0	63.5	75.0	72.0	—
Purpurin Sulphonate	81.5	68.6	57.0	74.5	73.0	60.0

[1] Dye used at concentration of 0.4 mM.

Table 4b. Uptake of dyes (μ-moles) by hydroxyapatite, calcium phosphate, calcium carbonate and fluorapatite, in relation to the surface areas of these compounds

Dye	*Hydroxyapatite*			*$CaHPO_4$*	*$CaCO_3$*	*Fluorapatite*
	68	*36*	*8*	*1.9*	*0.6*	*1.79 m^2/g*
Alcoholic						
Acid Alizarin Blue	—	39.6	—	6.5	6.5	—
Alizarin Cyanin Green[1]	85.0	18.0	13.0	5.0	3.0	—
Alizarin Cyanin 2R	—	16.0	—	5.5	3.5	—
Alizarin Red S[1]	54.0	17.4	14.0	13.0	10.0	—
Alizarin Viridin[1]	78.0	22.0	14.0	8.0	5.0	—
Amaranth	—	35.0	—	13.5	9.0	—
Anthracene Blue SWX	—	38.0	—	12.5	6.5	—
Azo Acid Blue[1]	66.0	17.0	15.0	14.0	21.0	—
Azo Bordeaux	—	14.4	—	1.5	1.5	0
Bordeaux Red	89.0	53.5	40.0	1.5	0	3.0
Calcium Fast Purple[1]	66.0	20.0	17.4	5.0	3.0	—
Chrome Violet[1]	—	46.0	—	4.0	3.0	—
Chromotrope[1]	—	34.0	—	8.0	4.0	—
Dianil Blue[1]	177.0	86.0	60.0	38.0	31.0	—
Fast Yellow	—	14.0	—	7.0	4.5	—
Geranine G[1]	160.0	79.0	59.0	12.0	5.0	—
Naphthalene Black 10B[1]	—	26.0	17.0	5.0	3.0	—
Orange G	57.0	17.5	11.5	4.0	2.5	—
Orseillin BB	—	57.5	—	11.0	2.5	—
Ponceau 2R	55.0	38.5	29.0	2.0	1.5	4.5

[1] Dye used at concentration of 0.4 mM.

To determine the degree of specificity of the dyes towards the apatite structure, comparisons were made of the dye uptake in hydroxyapatite, secondary calcium phosphate ($CaHPO_4.2H_2O$), calcium carbonate and in a few experiments, fluorapatite. As adsorption of dyes is a surface effect, the surface areas of these compounds were first determined. Both alcoholic and aqueous stains were used. Graphs were constructed in which the uptake of dye was related to the surface area of three hydroxyapatite samples. Extrapolation of the lines gave the dye uptake expected with the other compounds, assuming that these behaved like hydroxyapatite. This method afforded a simple way of assessing and contrasting the specificities of the dyes for hydroxyapatite. Four examples of the results obtained are shown in Fig. 2. The experimental results are listed in Table 4. A few discrepancies are seen between the results in Tables 3 and 4. Some of these are more apparent than real as the expression of results in the tables is not strictly comparable. In Table 4 the uptake of dye before any washing procedure is quoted. In the case of some dyes in Table 4 concentrations of 0.4 mM were used, whereas only concentrations of 0.2 mM were used in the earlier study (Table 3). In a few instances, genuine differences in the uptake of dye by hydroxyapatite (36 m^2/g) are present in the two sets of tables and can only be attributed to small variations in the experimental conditions.

In general, specificity of dye towards hydroxyapatite is a more characteristic finding in alcoholic than aqueous dye solutions. In alcohol, Alizarin Red S and Azo Acid Blue were the least specific and Ponceau 2R the most specific. With most of the dyes in aqueous solution,

the uptake by calcium phosphate and calcium carbonate was greater than that by hydroxyapatite when allowances were made for differences in surface area measurement. Acid Alizarin Blue SWR, Geranine G and Chrome Violet G were atypical. While considering specificity, it is of interest to compare the relative adsorption of five dyes by hydroxyapatite and 'cellophane'. In the dialysis studies outlined earlier it was found that some dyes in tris medium were strongly adsorbed to the dialysis membrane. A method for determining the uptake on the membrane was devised and the dyes ranked in order of their uptake (Table 5). The dissimilar ranking suggests that some specificity exists within these dyes for these adsorbing surfaces.

Table 5. Adsorption of 0.2 mM dyes in tris buffer on dialysis membrane and hydroxyapatite

Dye	*% dye removed from solution by surfaces*			
	Hydroxyapatite	*Ranking*	*Cellophane*	*Ranking*
Alizarin Red S	25.0	5	1.0	5
Dianil Blue	78	2	63.3	1
Geranine G	53.5	4	24.6	3
Orseillin BB	92	1	25.0	2
Purpurin Sulphonate	68.6	3	14.2	4

From the results in Table 4 and Fig. 2 it is seen that in most instances the uptake of dye was directly related to the surface area of hydroxyapatite. A poor linear response was obtained with some dyes in tris, in particular with Alizarin Red S. Similarly with Orseillin BB, which, though giving very variable results, gave no clear evidence of any relationship between the uptake of dye and surface area. This was attributed to the apparent formation of two coloured products during the exposure of sparingly soluble calcium compounds to this dye in tris buffer.

It will be recalled that one of the initial objectives of these studies was to decide if dyes were taken up preferentially by the protein or mineral fractions of dental tissues. Although the results so far presented show that synthetic hydroxyapatite is certainly capable of extracting dye from solution, they do not exclude the possibility that protein may also do this in powdered calcified tissue. An attempt was made to study this more directly.

Powdered dentine, separated from enamel by flotation in a bromoform–acetone mixture, was sieved through a 400 mesh screen and a sample of this powder was rendered anorganic by prolonged refluxing with ethylene diamine in a soxhlet apparatus. The powder was then repeatedly washed with alcohol and finally with acetone. It was finally dried in a vacuum desiccator. Failure of the Van Gieson stain to give the characteristic colour of normal dentine was taken as evidence for the removal of most of the protein.

It was consistently found that anorganic dentine was capable of extracting more dye from solution than normal dentine. This result was contrary to that predicted if the dyes were preferentially adsorbed to protein and was probably explained by the increased surface area of the hydroxyapatite crystals, unmasked, as it were, by removal of the diffuse organic material. It was thought necessary to obtain some estimates of the surface area by comparing the uptake of dyes by hydroxyapatite samples having different surface areas with anorganic and normal dentine. The results for two of the dyes studied are given in Fig. 3. The

uptake of alcoholic Geranine G by dentine and anorganic dentine corresponded to hydroxyapatite with surface areas of 15 and 64 m^2/g respectively and with alcoholic Orange G to 22 and 64 m^2/g. The lack of agreement between the dyes in estimating the surface area of normal dentine might suggest that Orange G, which is adsorbed to the greater extent, is staining both protein and hydroxyapatite, whereas Geranine G is more specific for the mineral phase. However, surface area measurements on these dentine samples by the B.E.T. nitrogen adsorption method gave values of 20.9 m^2/g for normal dentine and 74.1 m^2/g for anorganic dentine. The estimates on normal dentine are thus better with Orange G than Geranine G.

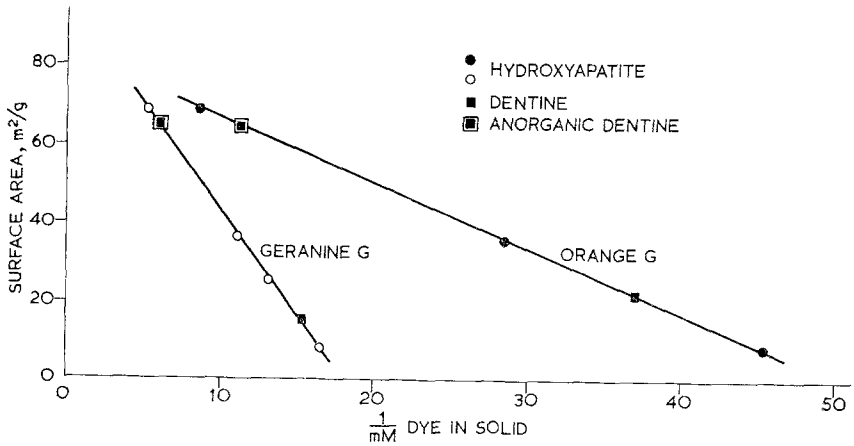


Figure 3. The uptake of dye solutions by hydroxyapatite, anorganic and normal dentine powder in relation to the surface areas of these samples.
○ hydroxyapatite samples exposed to 0.4 mM Geranine G in 95% ethanol; ● hydroxyapatite samples exposed to 0.4 mM Orange G in 95% ethanol; ■ dentine; ▣ anorganic dentine.

The question posed at the outset has therefore not been satisfactorily answered. These results do not rule out limited dye uptake by protein in normal dentine. Sections of decalcified bone certainly took up from alcoholic solutions Geranine G, Orange G (rather poorly), Orseillin BB and Naphthalene Black 10B. These stains are, therefore, not specific for hydroxyapatite. On the other hand, in the solvents and concentrations used in my studies, Alizarin Red S did not stain these decalcified bone sections. The reasonable agreement between estimates of surface area made by determining the adsorption of dyes on the one hand and conventional gas adsorption methods on the other justifies further use of alcoholic dye solutions to determine the surface area of samples of hydroxyapatite. However, the application of such a method to powdered calcified tissues would be limited by their heterogeneity, originating as it does from chemical and physical causes such as the presence of different forms of calcium phosphate, the presence of amorphous and crystalline material in varying ratios, presence of lipids and mucopolysaccharides, and other factors.

Discussion

Throughout this paper it has been suggested that dyes are adsorbed on to the surfaces of crystals. This might be criticized because the experimental conditions, in particular the short exposure times, possibly restrict the uptake of dyes mainly to the hydration shell or that

region close to the crystal surface where ions are relatively fixed. However, this academic distinction between crystal surface and hydration shell is difficult both to define and to study. In this discussion it is sufficient to realize that it is not known at which interface the dyes are adsorbed.

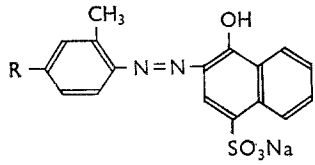
An analysis of the results relating the adsorption and retention of dyes to their chemical structure (Tables 1, 3 and 4) has provided some clues as to the groupings within the dye molecules which are responsible for the adsorption of dyes to hydroxyapatite. In considering this, the crystal surface of synthetic hydroxyapatite ought probably to be regarded as carrying a net negative charge, especially when placed in tris buffer at pH 7.4 and at the solid : solution ratios used in these studies (Neuman & Neuman, 1958; Leach, 1960; Napper & Smythe, 1966; Somasundaran, 1968). The ions in the hydration shell (*e.g.* H_3O^+) may balance this surface charge (Pak & Bartter, 1967). In 95% ethanol, the pH and distribution of charged ions are presumably different and these factors might be partly responsible for the contrasting behaviour of some dyes in tris and alcohol. For example, many of the dyes taken up by hydroxyapatite from alcoholic solution were readily desorbed (see Table 3b). This suggests that these dye molecules were held by weak Van der Waal's forces. An examination of the formulae and solubilities of such dyes did not reveal any feature which might be responsible for this behaviour.

Experimental studies by Myers (1964) have suggested that a chelate is formed between calcium of the hydroxyapatite surface and a bidentate ligand on the anthraquinonoid dye molecule consisting of the carbonyl oxygen and 1-hydroxyl group. This view is contrary to that of Kiel & Heertjes (1963) who thought that salt formation occurred through the 2-hydroxyl group. The possession of two ligands or active hydroxyl groups by Purpurin and Alizarin Bordeaux can be related to the degree of adsorption of these dyes by hydroxyapatite which was higher than in Alizarin and Anthrapurpurin, each of which has only one of these sites (Table 3a). This comparison can again be made between Purpurin Sulphonate with high uptake, and Alizarin Red S with only one site and much lower uptake. These dyes, like many others in aqueous alkaline solvents, showed little or no specificity towards the apatite structure and were adsorbed readily by other insoluble calcium salts. Myers noted this result also and gave this as evidence in support of the chelation theory of adsorption. However, sulphonation of Alizarin reduced the uptake of the dye and further substitution reduced it still more as in Anthracene Blue and Alizarin Cyanin 2R, despite these having greater potential to form chelates through having additional carbonyl and hydroxyl groups. From this and many other examples, it follows that there is no direct relationship between the total number of anionic groups in these anthraquinonoid dyes and their affinity for hydroxyapatite. The introduction of two basic amino groups in place of two hydroxyl groups distinguishes Acid Alizarin Blue GR, with very low uptake, from Anthracene Blue with a much higher uptake. Besides altering the overall charge on the dye, the amino groups reduce the number of chelating sites in the molecule. They might also indirectly affect the co-ordinating ability of the adjacent carbonyl oxygen atoms and activity of the 2-hydroxyl group.

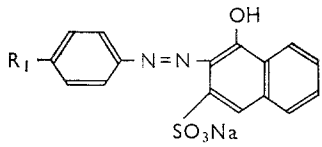
Another factor to be taken into account in considering the results is that of steric effects introduced by substitution of certain groups into dye molecules. These groups physically impede the formation of chelate rings or the adsorption on hydroxyapatite by other mechanisms, *e.g.* hydrogen bonding. It seems certain that Alizarin Viridin is an example of such an effect. The two large benzene rings form the arms of the Alizarin molecule and have considerably reduced the adsorption exhibited by this molecule. A control in this experiment is

provided by the behaviour of Alizarin Cyanin Green. From a comparison of the structures and uptake of these three dyes, it is concluded that the benzene rings prevent effective chelation or bonding of the carbonyl oxygen–hydroxyl unit to hydroxyapatite.

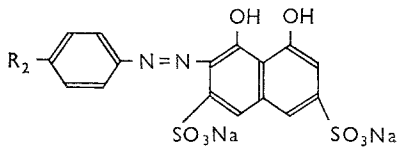
Turning now to the azo dyes in tris buffer and their adsorption by hydroxyapatite, several structural units are implicated in the adsorption of some of these dyes and it is of particular interest that three dyes which possess very similar groupings are adsorbed equally well in aqueous and alcoholic solvents. These groupings are found in Orseillin BB and Dianil Blue:



They have a slightly different arrangement in Geranine G:



The reduced uptake of Dianil Blue relative to Orseillin BB may be partly the result of hindrance created by the large terminal unit in Dianil Blue:



Theoretically, possession of two groups by Dianil Blue, each capable of binding to hydroxyapatite, should increase the ease of adsorption of the dye, although the number of dye molecules that can ultimately be adsorbed might be limited by hindrance created by one or other of these groups. At low concentrations of dye in solution, there would be a high affinity, but this would be reduced as the molecules had to jockey with each other for position on the crystal surface. Some evidence to support this view was obtained (see Table 2 and Fig. 2 in paper 2 of this series).

Minor structural variations of these units are found in several other dyes, *e.g.* Naphthalene Black, Azo Acid Blue, Sorbine Red and Ponceau 2R, but the uptake of these dyes from tris buffer was found to be small. It is not possible at this stage to decide if the reduced adsorption can be attributed to differences in these units or in other parts of the molecules. Although the same one unit may be responsible for the adsorption mechanism in these dyes, groups in more remote parts of the molecule must exert an influence on the activity of this unit. The nature of this effect could take several forms.

The most obvious reaction to account for the behaviour of some of these azo dyes is to suggest chelation between the azo groups, with dative covalency, and the adjacent hydroxyl group which forms a covalent link with calcium. Simply the presence of a hydroxyl group in an ortho position to the azo group cannot be the explanation since this arrangement is very common in dyes which show little affinity for hydroxyapatite in tris buffer. The ionization of these groups might be encouraged by the position of a neighbouring sulphonic acid

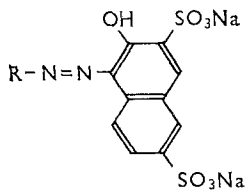
or methyl group and inhibited by others, e.g. amino and nitro groups. An examination of the structural formulae of the dyes and their adsorption ratings readily shows the complexity of the situation.

In the case of Geranine G, Orseillin BB and Dianil Blue, it seems that the particular mechanism governing their adsorption is influenced to only a minor degree by the nature of the solvent. Also in this category is Chrome Violet G, which showed an appreciable adsorption to hydroxyapatite in tris and in ethanol. The coloured compounds so produced were remarkable in their high degree of stability. This dye was included because of the possibility of chelate formation between the carbonyl oxygen of the dissociated sodium salt and the adjacent hydroxyl groups.

If chelation with calcium is the mechanism whereby these dyes are adsorbed from aqueous and alcoholic solvents, then it would follow that their degree of specificity for the apatite structure as distinct from other insoluble calcium salts should be small. It can be seen from Table 4 that the evidence is, if anything, against this view.

Certainly the simple anthraquinonoid dyes exerted little affinity for hydroxyapatite in alcohol, and chelation, if this is the mechanism of their union in aqueous solvents, seems to be suppressed in alcohol. Chelation cannot be ruled out, however, as an explanation for the adsorption of some dyes in alcohol. Somehow one has to account for the slightly higher uptake and retention of Alizarin Viridin compared with Alizarin Cyanin Green despite the finding that the simple parent Alizarin molecule was not adsorbed. Could the principal point of union in these two dyes (Alizarin Viridin and Alizarin Cyanin Green) be through the imine (—NH—) linkages? This might be via the imine and carbonyl oxygen of the quinonoid or alternatively the imine and carbonyl oxygen of the sulphonic acid, with the methyl groups in the benzene rings exerting a directing influence. The lack of a high degree of specificity of these anthraquinonoid dyes towards the apatite structure would support the view that a limited amount of chelation with calcium is possible even in alcohol.

Some striking comparisons can be made between the behaviour of the azo dyes in tris and alcohol. The functional unit in Orseillin BB, Geranine G and Dianil Blue has been shown to retain its activity in alcohol. A slight modification of this unit is found in Bordeaux Red, Amaranth and Ponceau 2R, which exhibit high degrees of adsorption and specificity for apatite in alcohol only. This unit:



or part of it, is presumably responsible for this marked difference in the two solvents. It is tempting to suggest that the precise positioning of the hydroxyl, sulphonate and azo linkage in the naphthalene nucleus is responsible for the adsorption of these dyes. Although the overall chemical formulae of Azo Bordeaux and Bordeaux Red are identical, the positioning of the groups in Azo Bordeaux is different and the uptake is correspondingly different. In Orange G there is further rearrangement of these groups, and in addition, a terminal phenyl ring replaces the naphthalene of Bordeaux Red. These alterations led to a marked reduction in the uptake of Orange G. As most of these azo dyes in alcohol showed a

pronounced preference for hydroxyapatite rather than fluorapatite or other calcium salts, chelation with calcium is unlikely to explain their adsorption.

Several mechanisms can therefore be considered in an attempt to account for the adsorption of dyes with particular structural features. Included in these are the following:

(1) chelation of surface calcium by bidentate ligands formed possibly from the functional groups —NH and C=O, —OH and C=O, or —N= and —OH. This occurs mainly in the tris medium.

(2) Hydrogen bonding between hydroxyl groups in hydroxyapatite and azo-nitrogen, —NH—or ionized carbonyl oxygen in the dye molecule. These hydroxyl groups in hydroxyapatite might even occur as part of the orthophosphate groups in calcium-deficient apatites (Bett, Christner & Hall, 1967).

(3) preferential adsorption of ethanol, perhaps by hydrogen bonding followed by chemisorption of dye on to the alcohol. This has been found on the surface of alumina and silica (Kipling, 1965), but it is not known to which atoms in hydroxyapatite ethanol might attach itself. Incidentally, Pak & Bartter (1967) claim that tris, which behaves as a cation at pH 7.4, was not adsorbed by synthetic hydroxyapatite.

(4) a replacement of surface (or hydration shell) phosphate by negatively charged groups in the dye molecule, particularly sulphonate ($—SO_3^-$), permitting attachment of the dye through these groups.

(5) there may be complex formation or adsorption in which the union resembles an acid–base type of reaction or salt formation, with negatively charged groups on dye molecules linking with positively charged surface calcium. In the anthraquinonoid dyes, there is evidence that union with calcium can occur at the 2-hydroxyl position (Kiel & Heertjes, 1963).

A series of experiments was planned to study some of these possible reactions and to provide an insight into the nature of the adsorption process and these will be considered in a subsequent paper.

A point of some practical importance is the apparent failure of several dyes used for the staining of bone sections to stain synthetic hydroxyapatite. Included in this category are Acridine Red, Biebrich Scarlet and Nile Blue Sulphate. Other dyes, for example Alizarin Cyanin Green and Fast Yellow, showed little adsorption to hydroxyapatite under the conditions of these experiments.

Studies are now in progress to evaluate the use of some of the high-affinity dyes in the staining of undecalcified sections of bone. The results of the preliminary studies have been discouraging in so far as there was poor correlation between the ability of some dyes to stain bone sections and hydroxyapatite.

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Addendum

Since the completion of this manuscript a comprehensive study on the staining mechanism of Alizarin and Alizarin Red S has been published by Puchtler, Meloan & Terry, *J. Histochem. Cytochem.* **17**, 110–24 (1969).

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